

Formation-Aware Event-Triggered Coordination for Vineyard Bird Deterrence

Abstract

This document specifies a real time, vineyard structured coordination framework for heterogeneous aerial and ground robots that deter birds. The approach converts multi sensor detections into a value weighted demand map on a row graph, partitions the field into robot zones, and reacts inside each zone with light assignments, formation rules, and contextual deterrence mode scheduling. Fault detection, health scoring, and graded derating drive resilient reallocation under intermittent communication. The framework targets small to medium fleets and emphasizes reliability, safety, and fast decisions suitable for field conditions.

1 Introduction and Motivation

California vineyards experience bursty, spatially concentrated bird activity that varies by hour, light level, and ripeness stage. Static deterrents lose effectiveness due to habituation. Growers need a mobile, adaptive system that can sense birds, decide quickly where to send agents, and apply the right cues without overusing any single mode.

The target deployment is a small team of unmanned aerial vehicles and unmanned ground vehicles operating over row structured geometry with headlands and no fly buffers. Communication may be intermittent. Compute resources on board or at the edge are limited. Operators need a system that behaves predictably, respects safety envelopes and regulations, and provides measurable reduction in crop loss with minimal overhead.

This work focuses on a coordination method that exploits the vineyard structure. Rather than solving large coupled problems, we partition the row graph into zones based on a value weighted demand field, then react within each zone using short horizon assignments, rule based formation selection, and a lightweight learner to schedule deterrence modes. The design integrates fault detection and fault tolerant behavior from the start.

2 Problem Statement and Assumptions

Environment and robots

Let the vineyard be represented by a row graph $G = (V, E)$ whose nodes are row segments and headlands, with edges that encode traversable connections and directions. Let R^A denote the set of aerial robots and R^G the set of ground robots, with $R = R^A \cup R^G$. Each robot $r \in R$ has kinematic limits, an energy budget, and a payload mode library for deterrence.

Demand and value

Perception produces a time varying intensity $\lambda(x, t) \geq 0$ on nodes or short arc length cells $x \in V$. A value weight $w(x, t) \geq 0$ encodes varietal value, phenology stage, proximity to field edges, and habituation memory. Planning uses the value weighted demand

$$q(x, t) = w(x, t) \lambda(x, t).$$

Events of interest are intercepts of predicted flock paths and satisfaction of patrol quotas that keep pressure low in high value zones.

Objective and constraints

Given $q(x, t)$, initial robot states, and safety and regulatory constraints, produce in real time:

1. a zone assignment that allocates disjoint row sets to robots and updates only on triggers,
2. intercept or patrol waypoints, formation parameters when needed, and deterrence modes for the next short horizon,
3. health aware reallocation and graceful degradation under faults and communication dropouts.

The objective is to reduce time integrated crop loss at risk while respecting separation, geofences, speed and energy limits, and workload limits for the operator. The system must remain responsive during bursts when detections spike and communication is degraded.

Assumptions

- The row graph and geofences are known and change slowly relative to decision cycles.
- Detections can be fused into short tracks with timestamps and confidences.
- The fleet size is small to medium and the number of simultaneous hot spots is limited.
- Formation maneuvers are executed by low level controllers with safety filters.

3 Contributions and Novelty

We introduce a vineyard structured, real time coordination stack for heterogeneous aerial and ground robots that deter birds by coupling detection, value mapping, zone partitioning, lightweight intercept logic, rule based formations, contextual mode scheduling, health driven reallocation, and a lean communication protocol. The key contributions are:

1. **Value at risk planning.** Decisions are driven by a value weighted demand field $q(x, t) = w(x, t)\lambda(x, t)$ that encodes varietal, phenology, edge exposure, and habituation history, aligning actions with expected crop loss rather than uniform coverage or raw detection intensity.

2. **Health aware zone partitioning on a row graph.** Zones are assigned with power weights that include geodesic distance, per robot energy, and a live health score. Contiguity repair and hysteresis deliver stable, connected ownership of rows and prevent thrashing at boundaries.
3. **Partition then react coordination with event triggers.** The system recomputes zones only when value mass, high value intercepts, link class, or health cross thresholds, and reacts inside zones with short horizon action scoring and tiny assignments. This yields predictable sub second decisions under bursty demand and intermittent links.
4. **Rule based formation selection with safety filtered execution.** Templates such as sweep, arc, parallel row screen, and UAV over UGV screen are selected by local geometry and wind rules, then executed by low level controllers with barrier function safety filters. Formation behavior is achieved without a global combinatorial layer.
5. **Contextual bandit for deterrence mode scheduling.** Mode rotation is learned online per row or flock using context from species, lighting, wind, recent mode history, and current formation to slow habituation while accounting for energy cost.
6. **Communication lean, event sourced protocol.** Coordination exchanges compact deltas with time to live and deterministic conflict resolution, allowing robots to continue locally during dropouts while keeping bandwidth low.
7. **Fault detection that drives coordination.** Normalized innovation tests, CUSUM or GLR drift checks, progress residuals, and energy or link monitors are fused into a graded health score that directly modulates zone ownership, speeds, payload use, and role substitution.
8. **Vineyard specific engineering.** The approach exploits row geometry, headlands, and flight buffers in zone logic, intercept scoring, and patrol quotas, improving practicality for field deployment.

3.1 Comparison to Prior Work

Centralized task routing and global planning. Existing approaches often rely on large coupled planners or frequent global replanning, which assume reliable connectivity and ample compute. Our method avoids large coupled solves at runtime, using event triggered partitioning and in zone reaction to preserve responsiveness during detection bursts and communication degradation.

Voronoi coverage and persistent monitoring. Classical coverage maximizes uniform presence or detection probability. We optimize a value weighted objective tied to crop economics, and add contiguity repair, hysteresis, and health driven offsets to stabilize zones and better match operational risk.

Auction and consensus based allocators. Market style methods can require sustained message exchange and consistent neighbor sets. Zone ownership with event sourced deltas minimizes traffic, while temporary zone merging still enables cooperation when adjacent hot spots arise.

Fixed cue schedules for wildlife deterrence. Many systems rely on static or manually tuned cue rotations. Our contextual bandit adapts mode selection to species and recent history, addressing habituation directly with online learning.

3.2 Scope and Limitations

The design prioritizes reliability and speed in small to medium fleets with intermittent links. During widespread, simultaneous spikes the lack of a global optimizer can leave some global efficiency unrealized. Bandit rewards assume quasi stationary response over short horizons and may require reset logic when species or conditions shift rapidly. Performance depends on the fidelity of detection tracks and the value map; degraded perception will bias partitions and action scores.

4 System Architecture

The stack contains five layers: perception and demand inference, formation and task abstractions, partition then react coordination, fault detection and health scoring, and execution and safety. Layers communicate through compact messages that carry deltas only.

Layer A. Perception and Demand Inference

- **Multi sensor ingestion.** Camera and acoustic detections with confidences and timestamps. Optional radar where available.
- **Tracking and aggregation.** Multi target tracking on the row graph to build short tracks and predict near term flock paths.
- **Intensity estimate.** Estimate $\lambda(x, t)$ with either a self exciting point process or a Gaussian process prior over the row graph. Maintain uncertainty for use in triggers and safety margins.
- **Value map.** Compute $w(x, t) = w_{\text{phen}}(t) w_{\text{var}}(x) w_{\text{edge}}(x) w_{\text{hab}}(x, t)$ where phenology and varietal terms reflect crop value, edge term reflects exposure, and habituation term decreases with recent cue usage and recovers with rest.
- **Planning field.** Set $q = w\lambda$. Smooth lightly across adjacent rows to avoid jitter, then clip to bounds.

Layer B. Formation Templates, Tasks, and Modes

- **Templates.** Sweep line across rows, arc around flock, parallel row screen, UAV screen over UGVs. Each template defines a coverage kernel, nominal speed, and role set.
- **Tasks.** Patrol tasks with coverage quotas by zone. Intercept tasks with time windows anchored to predicted flock path points. Escort tasks to shepherd flocks off field.
- **Modes.** Per robot library of deterrence modes with energy and efficacy models. Scheduling is handled by a contextual bandit.

Layer C. Partition then React Coordinator

C1. Zone partitioning

Problem. Assign each row cell $x \in V$ to an owning robot r to balance value mass and respect travel costs and energy headroom, with preference for contiguous zones.

Weights. For robot r at pose p_r and health score $h_r \in [0, 1]$, define a power weight

$$\pi_r(x) = \alpha D(p_r, x) + \beta E_r^{\text{unit}} + \gamma (1 - h_r),$$

where D is geodesic distance on the row graph, E_r^{unit} is marginal energy per meter for r , and α, β, γ are scale factors.

Assignment rule. Compute

$$\text{owner}(x) = \arg \min_{r \in R} (\pi_r(x) - \theta_r),$$

with offsets θ_r chosen to balance $\sum_{x \in V_r} q(x, t)$ across robots, where V_r is the zone of r . Offsets can be updated by a few iterations of proportional control to match a target value mass per robot. Ties break by stable robot identifiers to avoid oscillations.

Contiguity. After initial assignment, enforce contiguity by extracting connected components per robot and reassigning stray fragments to nearest neighbors that increase balance the least. This runs in near linear time in $|V|$.

Hysteresis. Maintain zone boundaries with hysteresis. A cell switches owner only if the new owner beats the current owner by a margin δ_{zone} . This reduces thrashing.

C2. Triggers for repartitioning

Recompute zones only when at least one condition holds:

- Total value mass per zone changes by more than τ_{mass} .
- A new intercept task with value above τ_{task} appears near a boundary.
- A health score crosses a threshold or a robot joins or leaves.
- Link quality changes class and neighboring handoff is needed.

Each trigger has a minimum hold time to prevent rapid toggling.

C3. Intercept and patrol inside a zone

For each zone owner r :

1. Predict near term intercept candidates by sampling future flock waypoints in the zone over a horizon H seconds.
2. Build a small assignment between a set of candidate actions a and the intercept candidates, or choose the top scoring action when there is at most one active flock.

3. Score actions by

$$S(a, r) = \sum_{c \in \mathcal{C}_H(a)} w_c \lambda_c g(a, c) - \alpha E(a) - \beta \text{Switch}(a) + \eta h_r,$$

where $g(a, c)$ is expected suppression gain in cell c , $E(a)$ is energy, and $\text{Switch}(a)$ penalizes formation or mode changes.

4. If two adjacent zones are hot and within cooperation distance, temporarily merge their cell sets and run the same procedure for the pair.
5. When no intercepts are present, execute patrol by visiting cells with largest q subject to a coverage quota and row direction constraints.

C4. Formation selection by rules

Select templates without optimization:

- Use arc around flock if predicted flock radius exceeds $r_{\min}$ and wind shear is below a limit.
- Use parallel row screen if two adjacent rows exceed a value threshold.
- Use UAV screen over UGV when UGV speed is the bottleneck and aerial presence improves response time.
- Revert to sweep for diffuse demand or when membership falls below required roles.

Membership is assigned by proximity and role priority. If a member derates, collapse to a simpler template until backup arrives.

C5. Contextual bandit for mode scheduling

For each row or active flock, run a contextual bandit with context ϕ that includes species, time of day, light, wind, recent mode history, and current formation. Arms are deterrence modes. Reward is the immediate reduction in q minus a small energy cost and a habituation penalty that increases with repeated use. Algorithms such as UCB or Thompson sampling are suitable. The bandit returns the mode for each robot and updates after each action.

C6. Communication protocol

Messages are event sourced deltas with time to live. Zone updates carry boundary hashes and a short list of changed cells. Intercept claims include task id, owner id, and expiry. Mode changes include robot id, mode, and reason code. Conflicts resolve by deterministic priority and timestamp. Robots cache the last known state and continue with local policies during dropouts.

Layer D. Fault Detection, Health Scoring, and Tolerance

D1. Residual based detection

- **State innovation tests.** Normalized innovation squared from the state estimator for each sensor channel. Flag when the statistic exceeds a chi square threshold for a dwell time.
- **Drift tests.** CUSUM or generalized likelihood ratio on gyros, motors, and altimeters to detect slow drifts and bias jumps.
- **Battery and link monitors.** Thresholds on state of charge, voltage sag under load, packet loss, and round trip time.
- **Progress residuals.** Compare expected task progress to realized progress over a short window and flag stalls.

D2. Health score and actions

Map detections to a scalar health score

$$h_r = \text{clip}\left(1 - \sum_k w_k s_{k,r}, 0, 1\right),$$

where $s_{k,r}$ are normalized statistics and w_k are weights. Smooth h_r with a short time constant. Use h_r to:

- derate speed, reduce aggressive formations, and limit energy intensive modes,
- downweight action scores and zone ownership priority,
- trigger handoffs and role substitutions when thresholds are crossed,
- enter safe policies such as hover or headland wait when h_r is low.

Layer E. Execution and Safety

High level outputs are waypoints, formation parameters, and mode commands. Low level execution uses existing controllers with safety filters such as control barrier functions to enforce separation, geofences, and row adherence. Interfaces follow ROS 2 conventions for command topics, status topics, and health hooks. Logs capture decisions, triggers, and outcomes for later analysis.

5 Algorithms and Pseudocode

Main cycle

1. Fuse detections and update λ . Update w and compute $q = w\lambda$.
2. If a trigger fires, recompute zones with power weighted assignment, contiguity repair, and hysteresis.
3. For each zone owner, build intercept candidates and select the next action by the score $S(a, r)$. If needed, apply a formation rule and role assignment.

4. Query the bandit for mode selection. Issue commands to Layer E.
5. Update residuals and health scores. If thresholds are crossed, adjust derating and request repartitioning if needed.
6. Transmit deltas. Repeat at the decision rate or on events.

Complexity and timing

Let $|V|$ be the number of row cells and $k = |R|$. One partition pass is near linear in $|V|$ with a small constant factor and a few balancing iterations. Intercept selection operates on tens of candidates per zone, which is millisecond scale on embedded CPUs. Bandit updates and health filters are constant time. The design targets sub second decisions during bursts.

6 Experimental Plan

Simulation

- Vineyard block with row geometry, buffers, wind, and terrain options.
- Demand from flock dynamics with shocks and optional playback of detections.
- Coordinators tested: partition then react and a hybrid that allows temporary zone merging.
- Baselines: greedy independent agents, fixed rate patrol, and simple nearest intercept without value weights.
- Metrics: time to intercept, percent value weighted intensity suppressed, energy per suppressed event, compute time, communication volume, formation and mode switching counts, and safety violations.
- Ablations: remove value weights, replace bandit with round robin, remove hysteresis, disable health weighting, vary trigger thresholds.

Pilot Field Trials

Two ground robots and two to three aerial robots at a test vineyard during harvest nights. Ground truth with stationary cameras. Record deterrence events, response times, operator workload, link quality, and stability of zone boundaries.

7 Implementation Notes and Defaults

- Decision cycle 0.5 to 1.0 seconds. Partition trigger hold time 5 to 10 seconds.
- Zone hysteresis margin δ_{zone} set to the 75th percentile of current $|\pi_{r^*}(x) - \pi_r(x)|$ across boundary cells.
- Health thresholds at 0.7 and 0.4 for mild and severe derating.
- Bandit exploration parameter selected to keep exploration under 15 percent of actions during peak harvest.
- Patrol quotas scaled to maintain a minimum dwell time in the top 20 percent of q cells per zone.

8 Risks and Mitigations

- Sparse or biased detections: fuse acoustic corroboration, widen prediction covariance, and fall back to patrol quotas in low confidence regions.
- Compute limits: keep horizons short, cap candidate action counts, and use cached shortest paths on the row graph.
- Communication dropouts: rely on zone ownership and local caches. Messages carry deltas with time to live. Robots continue with local policies until links recover.
- Regulatory limits: respect geofences and line of sight procedures. Log all formation and mode changes with time and reason code.

Appendix A. Notation

Symbol	Meaning
R^A, R^G	sets of aerial and ground robots
$G = (V, E)$	row graph with nodes V and edges E
$\lambda(x, t)$	demand intensity on node or cell x
$w(x, t)$	value weight by phenology, varietal, edge risk, habituation
$q(x, t)$	value weighted demand $w\lambda$
h_r	health score in $[0, 1]$
$S(a, r)$	action score for selection inside a zone

Appendix B. Message types

- Zone update: {boundary hash, changed cells, timestamp}
- Intercept claim: {task id, owner id, expiry, timestamp}
- Mode change: {robot id, mode id, reason, timestamp}
- Health update: {robot id, h_r , derating flags, timestamp}

References

- [1] J. Cortés, S. Martínez, T. Karatas, and F. Bullo, “Coverage control for mobile sensing networks,” *IEEE Transactions on Automatic Control*, vol. 49, no. 8, pp. 1327–1332, 2004.
- [2] F. Aurenhammer, “Power diagrams: properties, algorithms and applications,” *SIAM Journal on Computing*, vol. 16, no. 1, pp. 78–96, 1987.
- [3] H. W. Kuhn, “The Hungarian method for the assignment problem,” *Naval Research Logistics Quarterly*, vol. 2, no. 1–2, pp. 83–97, 1955.
- [4] P. Tabuada, “Event triggered real time scheduling of stabilizing control tasks,” *IEEE Transactions on Automatic Control*, vol. 52, no. 9, pp. 1680–1685, 2007.
- [5] C. E. Rasmussen and C. K. I. Williams, *Gaussian Processes for Machine Learning*. Cambridge, MA: MIT Press, 2006.

- [6] A. Reinhart, “A review of self exciting spatio temporal point processes and their applications,” *Statistical Science*, vol. 33, no. 3, pp. 299–318, 2018.
- [7] A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, “Control barrier function based quadratic programs for safety critical systems,” in *Proceedings of the IEEE Conference on Decision and Control*, 2014.
- [8] Y. Bar Shalom, X. R. Li, and T. Kirubarajan, *Estimation with Applications to Tracking and Navigation*. New York, NY: John Wiley and Sons, 2001.
- [9] A. Slivkins, “Introduction to multi armed bandits,” *Foundations and Trends in Machine Learning*, vol. 12, no. 1–2, pp. 1–286, 2019.
- [10] K. K. Oh, M. C. Park, and H. S. Ahn, “A survey of multi agent formation control,” *Automatica*, vol. 53, pp. 424–440, 2015.